WORKED SOLUTION

STEP III 2005, Question 11

Mechanics • Rigid Body Equilibrium

STEP III 2005, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q11

The key idea

The spindle and rods form a rigid body turning about a fixed horizontal axis, so one angle θ describes everything and energy is the natural tool. The potential energy is a sum of three cosines whose angles differ by $\frac{2}{3}\pi$; combining them gives a single expression $A\cos\theta + B\sin\theta$, which is one cosine wave in disguise. Once it's written in that form, the equilibria, their stability and the energy needed to get over the top can all be read off.

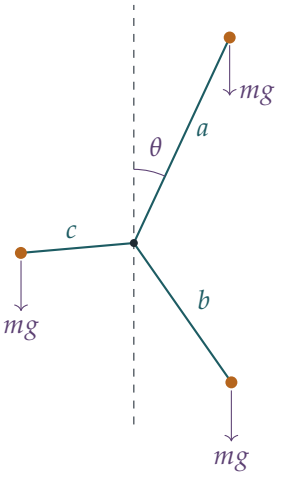
Solution

The energy

We model the particles as point masses on light rigid rods, with a smooth bearing so no energy is lost. Measure every angle clockwise from the upward vertical, as in the diagram. The rods of lengths a, b, c then make angles θ, $\theta + \frac{2}{3}\pi$ and $\theta + \frac{4}{3}\pi$ with the upward vertical, so the particles are at heights

$$a\cos\theta, \qquad b\cos\left(\theta + \tfrac{2}{3}\pi\right), \qquad c\cos\left(\theta + \tfrac{4}{3}\pi\right)$$

above the spindle.



Using $\cos\frac{2}{3}\pi = \cos\frac{4}{3}\pi = -\frac{1}{2}$, $\sin\frac{2}{3}\pi = \frac{\sqrt{3}}{2}$ and $\sin\frac{4}{3}\pi = -\frac{\sqrt{3}}{2}$,

$$\cos\left(\theta + \tfrac{2}{3}\pi\right) = -\tfrac{1}{2}\cos\theta - \tfrac{\sqrt{3}}{2}\sin\theta, \qquad \cos\left(\theta + \tfrac{4}{3}\pi\right) = -\tfrac{1}{2}\cos\theta + \tfrac{\sqrt{3}}{2}\sin\theta.$$

Taking the spindle as the zero of potential energy,

$$\begin{aligned} V &= mg\left[a\cos\theta + b\cos\left(\theta + \tfrac{2}{3}\pi\right) + c\cos\left(\theta + \tfrac{4}{3}\pi\right)\right] \\ &= mg\left[\tfrac{1}{2}(2a - b - c)\cos\theta - \tfrac{\sqrt{3}}{2}(b - c)\sin\theta\right]. \end{aligned}$$

Each particle moves on a circle about the spindle with angular velocity $\dot\theta$, so the speeds are $a\dot\theta$, $b\dot\theta$, $c\dot\theta$. The rods are light, so the total energy is

$$E = \tfrac{1}{2}m\left(a^2 + b^2 + c^2\right)\dot\theta^2 + \tfrac{1}{2}mg\left[(2a - b - c)\cos\theta - \sqrt{3}\,(b - c)\sin\theta\right].$$

Equilibrium

The system is in equilibrium where $\mathrm{d}V/\mathrm{d}\theta = 0$, that is,

$$-\tfrac{1}{2}mg\left[(2a - b - c)\sin\theta + \sqrt{3}\,(b - c)\cos\theta\right] = 0.$$

Since $a > b$ and $a > c$, we have $2a - b - c > 0$. So $\cos\theta \neq 0$ at an equilibrium: if $\cos\theta = 0$ then $\sin\theta = \pm 1$ and the bracket would be $\pm(2a - b - c) \neq 0$. Dividing by $\cos\theta$,

$$\tan\theta = -\frac{(b - c)\sqrt{3}}{2a - b - c},$$

as required. The argument also runs backwards: any θ satisfying this equation makes $\mathrm{d}V/\mathrm{d}\theta = 0$. So the equilibria are exactly the solutions of this equation.

Exactly two equilibria, and which is stable

The tangent function has period π and takes each real value exactly once on each interval of length π where it is defined. So the equation has exactly two solutions in $0 \le \theta < 2\pi$, which differ by π. These are the two equilibrium positions.

To decide stability, write V as a single cosine. Put

$$A = \tfrac{1}{2}(2a - b - c), \qquad B = -\tfrac{\sqrt{3}}{2}(b - c),$$

so $V = mg(A\cos\theta + B\sin\theta)$. Then

$$\begin{aligned} 4(A^2 + B^2) &= (2a - b - c)^2 + 3(b - c)^2 \\ &= 4a^2 + 4b^2 + 4c^2 - 4ab - 4bc - 4ca \\ &= 2\left[(a - b)^2 + (b - c)^2 + (c - a)^2\right] = 4R^2, \end{aligned}$$

so $A^2 + B^2 = R^2$, with $R > 0$ since the lengths are distinct. Choose α with $\cos\alpha = A/R$ and $\sin\alpha = B/R$. Since $A > 0$ and $B < 0$, $-\frac{1}{2}\pi < \alpha < 0$. Then

$$V = mgR\cos(\theta - \alpha).$$

The two equilibria are $\theta = \alpha$, where $V = mgR$ is a maximum, and $\theta = \alpha + \pi$, where $V = -mgR$ is a minimum.

? **Ask yourself** Why does a minimum of V mean stable?

Here $V''(\theta) = -mgR\cos(\theta - \alpha)$. At $\theta = \alpha + \pi$ this is $mgR > 0$, so a small displacement raises the potential energy. With total energy fixed, the system can't wander far, and it oscillates about this position. At $\theta = \alpha$ we have $V'' = -mgR < 0$: a small displacement lowers V, the kinetic energy grows, and the system moves away. So the stable equilibrium is $\theta = \alpha + \pi$, the one with $\frac{1}{2}\pi < \theta < \pi$, where $\cos\theta < 0$ and $\sin\theta > 0$. In that position the longest rod a points below the horizontal, at an angle $|\alpha|$ to the downward vertical. The other equilibrium, with $-\frac{1}{2}\pi < \theta < 0$ and rod a above the horizontal, is unstable.

Complete revolutions

Energy is conserved, since the bearing is smooth and gravity is conservative. Write $I = m(a^2 + b^2 + c^2)$, so that

$$\tfrac{1}{2}I\dot{\theta}^2 + mgR\cos(\theta - \alpha) = E.$$

To make complete revolutions the system must get past every angle, and in particular past $\theta = \alpha$, where V takes its largest value mgR. Since $\frac{1}{2}I\dot{\theta}^2 \ge 0$ everywhere, this needs $E \ge mgR$. Let ω be the angular velocity when the system passes through the stable position $\theta = \alpha + \pi$, where $V = -mgR$. Then $E = \frac{1}{2}I\omega^2 - mgR$, and the condition becomes

$$\tfrac{1}{2}I\omega^2 - mgR \ge mgR \quad \Longleftrightarrow \quad \omega^2 \ge \frac{4mgR}{I} = \frac{4gR}{a^2 + b^2 + c^2}.$$

So the system must pass through its stable equilibrium with angular velocity at least $\sqrt{4gR/(a^2+b^2+c^2)}$, as required.

(In the borderline case of equality, the system creeps towards the unstable equilibrium and only reaches it in infinite time. Strictly, a full revolution needs ω strictly greater than this value; the question asks only for the necessary bound.)

Checks

- If $a = b = c$ then $R = 0$ and $V \equiv 0$: the three equal masses balance in every position (their centre of mass is on the axis), and any spin at all goes round. This matches the bound tending to zero.
- If $b = c$, the tangent condition gives $\tan\theta = 0$. By symmetry, the long rod points straight up (unstable) or straight down (stable).
- A numerical check confirms the form of V, the identity $A^2 + B^2 = R^2$, and that there are exactly two equilibria with the stated stability, for random lengths. For $a = 3$, $b = 2$, $c = 0.5$ and $g = 9.81$, the critical speed is 2.5406 rad s^{-1}. Stepping the equation of motion shows that 2% below this the system swings back, while 2% above it goes right round.

Common slips

- Putting the rods at $\theta - \frac{2}{3}\pi$ and $\theta + \frac{2}{3}\pi$ the wrong way round relative to the diagram flips the sign of the $\sin\theta$ term, and with it the sign in the tangent condition.
- Concluding "two solutions" without noting that $\cos\theta = 0$ can't be an equilibrium, so that dividing by $\cos\theta$ loses nothing.
- Using kinetic energy $\frac{1}{2}m(a+b+c)^2\dot{\theta}^2$ or similar. Each particle has its own speed $r\dot{\theta}$, so the moment of inertia is $m(a^2+b^2+c^2)$.

Beyond STEP

The system is a compound pendulum, a rigid body about a fixed axis. Its energy depends only on the moment of inertia and on the position of the centre of mass, which lies at distance $R/3$ from the axis here. The picture of $\frac{1}{2}I\dot{\theta}^2 + V(\theta) = E$ as a curve in the $(\theta, \dot{\theta})$ phase plane, with oscillations inside a separatrix and rotations outside it, is a central idea in first-year dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body rotation and energy methods for this kind of problem.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.